



Experiment 4

NAME: Vedant Naik

ROLLNO: 41

CLASS: SE-2

Title

Develop a dynamic webpage where content updates on button click using JavaScript DOM. Implement jQuery effects: hide/show elements, toggle visibility, and animate elements.

Theory:

1. JavaScript and the DOM (Document Object Model):

- The **DOM** is a hierarchical representation of the elements on a webpage. It allows programming languages like JavaScript to interact with the structure, style, and content of the page.
- Using **DOM methods**, we can:
 - Access elements (getElementById, getElementsByClassName, querySelector)
 - Change content (innerHTML, textContent)
 - Modify attributes (setAttribute, removeAttribute)
 - Change styles dynamically (style.property)
- Example: Changing the text of a paragraph when a button is clicked enhances user interactivity.

2. Dynamic Webpage Content:

- A **dynamic webpage** changes content without reloading the page.
- Dynamic updates improve **user experience** by allowing real-time feedback and interactivity.
- In this lab, content updates are achieved using document.getElementById().innerHTML to manipulate text dynamically.
-

3. Introduction to jQuery:



- **jQuery** is a lightweight, fast JavaScript library that simplifies:
 - HTML document traversal and manipulation
 - Event handling
 - Animations and effects
- jQuery is **cross-browser compatible**, meaning the same code works in different browsers without modification.

Syntax Example:

`$(selector).action();`

- `$()` is the jQuery function
- `selector` identifies the HTML element
- `action()` is the method applied (like `hide()`, `show()`, `animate()`)

4. jQuery Effects and Animations:

- **Hide/Show:** Instantly hide or show elements.
- **Toggle:** Switch visibility state of an element (hide → show, show → hide).
- **Animate:** Allows smooth transitions of CSS properties (like height, width, opacity, margins).
- **Benefits of jQuery Effects:**
 - Improves visual interactivity
 - Reduces manual CSS manipulation
 - Provides smoother user experience

5. Integration of JS DOM and jQuery:

- Using **pure JavaScript** allows control over element content and structure.
- Using **jQuery** complements JS by handling animations and effects more efficiently.
- Combining both allows for **dynamic and interactive web pages** where content changes and user interactions are smooth and visually appealing.

6. Applications:

- Dynamic webpages are used in:
 - Online forms that validate inputs instantly
 - Content sliders and carousels
 - Real-time notifications



Vidyavardhini's College of Engineering and Technology, Vasai

Department of Computer Science & Engineering (Data Science)

■ Interactive dashboards

Develop a dynamic webpage where content updates on button click using JavaScript DOM. Implement jQuery effects: hide/show elements, toggle visibility, and animate elements.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Dynamic Webpage with jQuery Effects</title>

  <!-- jQuery CDN -->
  <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>

  <style>
    body {
      font-family: Arial;
      text-align: center;
      margin-top: 50px;
    }

    #box {
      width: 150px;
      height: 150px;
      background-color: coral;
      margin: 20px auto;
      position: relative;
    }

    button {
      margin: 5px;
      padding: 10px 15px;
    }
  </style>
</head>
<body>
```



Vidyavardhini's College of Engineering and Technology, Vasai

Department of Computer Science & Engineering (Data Science)

```
<h2 id="text">Hello! Click a button to change me.</h2>
```

```
<!-- DOM Update Button -->
```

```
<button onclick="changeText()">Change Text</button>
```

```
<!-- jQuery Effect Buttons -->
```

```
<br><br>
```

```
<button id="hideBtn">Hide</button>
```

```
<button id="showBtn">Show</button>
```

```
<button id="toggleBtn">Toggle</button>
```

```
<button id="animateBtn">Animate</button>
```

```
<!-- Box for animation -->
```

```
<div id="box"></div>
```

```
<script>
```

```
// ===== JavaScript DOM Manipulation =====
```

```
function changeText() {
```

```
    document.getElementById("text").innerHTML = "Text updated dynamically!";
```

```
}
```

```
// ===== jQuery Effects =====
```

```
$(document).ready(function () {
```

```
    // Hide
```

```
    $("#hideBtn").click(function () {
```

```
        $("#box").hide(1000);
```

```
    });
```

```
    // Show
```

```
    $("#showBtn").click(function () {
```

```
        $("#box").show(1000);
```

```
    });
```

```
    // Toggle
```

```
    $("#toggleBtn").click(function () {
```

```
        $("#box").toggle(1000);
```

```
    });
```

```
    // Animate
```

```
    $("#animateBtn").click(function () {
```

```
        $("#box").animate({
```

```
            left: '200px',
```

```
            height: '200px',
```

```
            width: '200px',
```

```
            opacity: '0.5'
```

```
        }, 1000).animate({
```

```
            left: '0px',
```



Vidyavardhini's College of Engineering and Technology, Vasai

Department of Computer Science & Engineering (Data Science)

```
height: '150px',  
width: '150px',  
opacity: '1'  
}, 1000);  
});
```

```
});  
</script>
```

```
</body>  
</html>
```

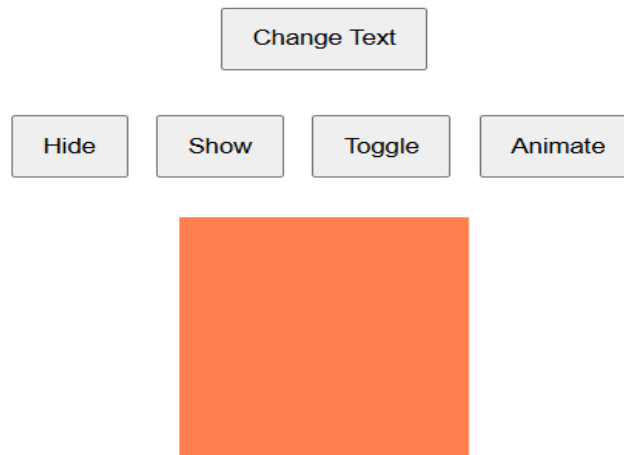


Vidyavardhini's College of Engineering and Technology, Vasai

Department of Computer Science & Engineering (Data Science)

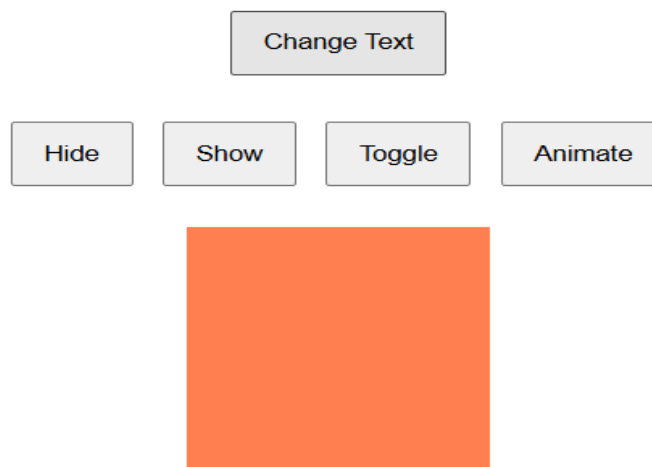
Output:

Hello! Click a button to change me.



#TEXT UPDATED

Text updated dynamically!





Vidyavardhini's College of Engineering and Technology, Vasai

Department of Computer Science & Engineering (Data Science)

#HIDE

Text updated dynamically!

Change Text

Hide

Show

Toggle

Animate

#SHOW

Text updated dynamically!

Change Text

Hide

Show

Toggle

Animate





Vidyavardhini's College of Engineering and Technology, Vasai

Department of Computer Science & Engineering (Data Science)

#TOGGLE

Text updated dynamically!

Change Text

Hide

Show

Toggle

Animate

Text updated dynamically!

Change Text

Hide

Show

Toggle

Animate





Vidyavardhini's College of Engineering and Technology, Vasai

Department of Computer Science & Engineering (Data Science)

#ANIMATE

Text updated dynamically!

Change Text

Hide

Show

Toggle

Animate





Conclusion

JavaScript DOM allows dynamic updating of webpage content in real-time. jQuery simplifies element manipulation, effects, and animations. Combining both creates interactive and visually appealing webpages, enhancing user experience. This lab demonstrated practical implementation of content updates, visibility toggling, and animations.

Question:

1. What is the difference between `.hide()` and `.toggle()` in jQuery?

The difference lies in **state management**:

- **.hide():** Specifically sets an element to display: none. If the element is already hidden, it does nothing.
- **.toggle():** Checks the current state of the element. If it is visible, it hides it; if it is hidden, it shows it. It essentially "flips the switch."

2. How can you stop an ongoing animation in jQuery before it completes?

To stop an animation before it finishes, you use the **.stop()** method.

By default, calling `.stop()` immediately clears the current animation but allows the next animation in the "queue" to start. You can pass two optional booleans: `stop(clearQueue, jumpToEnd)`.

3. How can you use jQuery to fade in and fade out an element?

jQuery provides built-in methods to handle opacity transitions over time.

- **.fadeIn():** Gradually changes the opacity from 0 to 1 (hidden to visible).
- **.fadeOut():** Gradually changes the opacity from 1 to 0 (visible to hidden).

.fadeToggle():